ChartScribe: Chart-to-Text Summarisation

Experiential Learning Project Report

Group members: [add names and roll numbers]

1. Problem Statement and Objectives

ChartScribe converts bar charts, pie charts, line charts, and infographics into structured data and concise natural-language summaries.

2. Methodology and Processing Pipeline

Image validation and resizing -> RapidOCR -> chart classification -> chart-specific geometry extraction -> insight analysis -> grounded summary.

3. Models and Tools

RapidOCR, OpenCV, NumPy, pandas, scikit-image, Streamlit, and optional Hugging Face google/flan-t5-small.

4. Implementation and Usability

Run pip install -r requirements.txt, pip install -e ., then streamlit run app.py. The CLI supports single-image and batch analysis.

5. Results and Limitations

The supplied sample charts demonstrate extraction, summaries, JSON export, and debug overlays. Small labels, blur, photographs, and unusual layouts can reduce accuracy.

6. Conclusion and Future Scope

The prototype provides an interpretable chart-to-text pipeline. Future work includes multi-series charts, multilingual OCR, perspective correction, and larger learned datasets.

7. Repository and References

GitHub URL: [add URL]. Video URL: [add URL]. References: RapidOCR, OpenCV, Streamlit, and Hugging Face FLAN-T5.